

OpenDDA Customer Release Guide

**DD04-300
Release 300**

OpenDDA

**OpenDDA
Customer Release Guide**

**DD04-300
Release 300
12/98**

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1. Introduction

1.1 About This Publication

Purpose

The OpenDDA Customer Release Guide provides

- details on functionality included in this release of Open Data Definition and Access (OpenDDA),
- minimum hardware and firmware revisions and minimum memory sizes,
- any special considerations when using the product,
- directions for installation of the OpenDDA software,
- problem reporting procedures, should any problems occur during the installation, migration, or ongoing use of OpenDDA.

Intended audience

This guide is intended for the system or network administrator's use to properly install OpenDDA.

Release

OpenDDA Release 300 is compatible with A^xM Software Release 300 and later.

1.2 Overview

Product description

OpenDDA (Open Data Definition and Access) 300 is a platform- independent software package providing a standard application programming interface. OpenDDA assists in developing and implementing applications with the following qualities:

- perform efficiently,
- read and write varying amounts of heterogeneous data,
- access one or more external data sources,
- can dynamically modify references to external data,
- can migrate, and
- are portable.

Platform

OpenDDA executes on the Application Modulex (A^xM) providing an open environment for the development and execution of custom control applications created by the customer.

1.3 OpenDDA Functionality

Features

The following table outlines the features of OpenDDA 300.

Table 1-1 OpenDDA Features

Feature	Description
Shared Libraries	Library files have changed from access type to shared type, which conserve disk and virtual memory and facilitate migration.
Remapping CIO files	This function allows an application to read into memory one or more CIO files, be able to select between them, clear one or more CIO files from memory, and write out a selected CIO to disk.
CIO File signature	A signature is bound to every CIO file created and a corresponding one to the associated executable application. When an application attempts to read in a new CIO file at run time, the signature is checked to make certain that it is a valid CIO file for that application. If it is not valid, an error is returned to the application and the offending CIO file is not read into the application.
LCN type ENTITY supported	An OpenDDA application can both read and write to a TDC 3000 ^x parameter of type ENTITY. This allows an application to interact with generic programs in the classic AM.
Dynamic Selection Sets	This function allows an application to modify the data source point name, parameter name, or array index at run time. This feature allows the programmer to create generic and repeatable programs, reducing development time and associated costs.
Hibernation	Allows a CL initiated application to initialize and then hibernate (suspend execution). CL initiated events can activate and terminate a hibernating application.
Inconsistency Error Enhancement	An INCONSISTENCY directive is added to the DEF file. It can specify ABORT or RETURN, which determines the behavior of the system when an inconsistency error is detected.
Cache	A cache of TDC 3000 ^x point name references is provided to facilitate superior performance when using Dynamic Selection Sets. A cache Maintenance Tool is provided to maintain the cache.
Array Access Performance Improvement	Data access time has been reduced for reading arrays of greater than 250 elements for all data types except strings and enumerations.
Licensing	This version of OpenDDA contains license software for accessing the OpenDDA executable. YOU MUST FOLLOW THE STEPS TO OBTAIN AND INSTALL A LICENSE KEY BEFORE LOADING OPENDDA. Refer to the License <i>Installation and Administration A^xM</i> manual.

Changed functionality for R200 and later

The default value of the Definition (.def) file qualifier STORE_CODE has been changed. The new default is NOSTORE. (The previous value in R100 and R110 was STORE.)

The application's .def file is no longer required at run time; however, in order to change any directives or qualifiers in the .def file, you are still required to have one. If no .def file is present, OpenDDA will attempt to use a system defined .def file in the /opt/DDA/bin directory. Otherwise, OpenDDA default values are used. Please refer to subsection 8.1 of the *OpenDDA Reference Manual* for more information.

Array access performance improvement

The array access performance enhancement in OpenDDA R300 is transparent to the user, other than when monitoring the performance of an application. There are no special functions to call or settings to make to take advantage of this enhancement.

When OpenDDA determines that an array of greater than 50 elements is being accessed for all data types other than strings and enumerations, it uses a more efficient internal call to do the data access.

Obsolete files

The following files are no longer used in OpenDDA (R200 and later):

/opt/DDA/bin/fortran.def

/opt/DDA/bin/c.def

/opt/DDA/lib/dda_lib.a

/opt/DDA/lib/dda_gdx_lib.a

/opt/DDA/lib/dda_gen_lib.a

Default CIO and DEF file names

Beginning with R200, the default CIO and DEF files that OpenDDA looks for are named by appending .cio and .def to the *executable* file name. For example, when you parse myapp.f, the files myapp.cio and myapp.def are generated. In R100 and R110, these are the default CIO and DEF files, regardless of the name of the executable. In R200 and R300, if you renamed the “myapp” executable to “demo1”, the default CIO and DEF files would be demo1.cio and demo1.def, instead of myapp.cio and myapp.def.

Precompiled source file name change

For release R100 of OpenDDA, the name of the precompiled FORTRAN source code generated by OpenDDA was application_name.src.f. For releases R110, R200, and R300 the file name has been changed to application_name_src.f. You should use the new build script and make files. If you have created any custom build scripts or make files, you will need to make the appropriate changes.

Inconsistency error for INTEGER*2

OpenDDA does not support statuses or values declared as INTEGER*2 or short int. The application will receive an inconsistency error if your HLL variables used with OpenDDA are declared as such.

1.4 Supporting Documentation

Documentation

Before installing and operating OpenDDA 300, you will need to refer to the following manual:

License Installation and Administration A^xM

In support of OpenDDA 300, the following documents have been published:

OpenDDA User's Guide

OpenDDA Reference Manual

When OpenDDA is installed on an A^xM, the following may also be useful:

Application Module^x Customer Release Guide

Application Module^x System Administration Manual

1.5 Software Package

Software media

OpenDDA 300 is distributed as a Software Development Kit on a 4 mm DAT tape.

Installing OpenDDA

After upgrading to HP-UX 10.20, you may use the HP-UX Software Distributor product to install your OpenDDA software. OpenDDA software on the tape can be loaded into an A^XM or workstation running HP-UX 10.20 that has a local DAT drive by using the `swinstall` command. It can be loaded into a Software Depot server by using the `swcopy` command.

Additional information on the installation procedure is provided in Section 3 of this guide.

2. Before You Install

2.1 Special Considerations

Load license software

Prior to loading OpenDDA R300, you must follow the following steps to set up the software licensing. Please refer the *License Installation and Administration A^xM* manual for detailed procedures for each step.

- Load Honeywell License Manager software
- Obtain license keys
- Edit the license file
- Set up the license environment variable for all OpenDDA users
- Determine if an options file is required, and, if so, set it up
- Install the license servers

Software loading order

To migrate your A^xM to OpenDDA R300, it is important to load software in the following order:

Table 2-1 Software Loading Order

Step	Software to Load
1	HP-UX 10.20 upgrade tape
3	Honeywell License software
4	A ^x M HP-UX software
5	A ^x M R300 personality
6	OpenDDA R300 software

Load compilers

Prior to using OpenDDA, the system administrator for your A*M or HP-UX system should load the compilers that you plan to use—the Series 700 HP-UX FORTRAN compiler, and/or the HP-UX ANSI C or C++ compiler.

Even though a compiler is not needed for the installation, it is required for building any OpenDDA application on the system.

Default NG

It is highly recommended that you assign a default NG System ID to your TPS system if one is not already defined. The OpenDDA cache bases its Entity names search on the NG as well as the point and parameter names. If the NG ID has not been assigned when OpenDDA begins to use the cache, and it is assigned later, there may be duplicate entries in the cache.

2.2 Hardware Preparation

Installing OpenDDA on a remote node

When developing your OpenDDA applications on a remote node, the minimum level software version of your HP-UX 700 or 800 series should be 10.20 (you can determine this with the command `uname -a`). The processor should be PA-RISC version 1.1 (you can determine this with the `getcontext` command). The system should be equipped with at least 32 MB (Megabytes) of memory. You can verify the amount of memory you have with the `dmesg` command (you must be logged in as root).

Application Module^x configuration

To determine the minimum hardware and software requirements for your A^xM, please consult the *Application Module^x Customer Release Guide*.

3. Installing OpenDDA 300

3.1 Overview

In this section

This section details the procedures for loading OpenDDA 300 from:

- a tape drive onto a software depot, or
- a tape drive onto the local HP-UX node, and
- a software depot to the Application Module^x.

swcopy command

The `swcopy` command prepares a device as a server.

You select filesets from the distribution tape and `swcopy` loads them into a target software depot.

swinstall command

You use the `swinstall` command to load filesets/partitions from the software source (may be a tape or a software depot) onto the system to be updated.

Removing filesets from the server

If, at a later time, you need to remove filesets from the software depot, you can unregister the depot and remove the filesets by using the `swremove` command. See the man pages for further information on the `swremove` command.

References

Refer to the *man* pages for `swremove`, `swcopy` and `swinstall` for more information on these commands.

See also the *Application Module^x System Administration* and the *Application Module^x Customer Release Guide* for additional information on installing software and backing up OpenDDA files and directories.

3.2 Copying OpenDDA to a Software Depot

Introduction

The following procedure describes the procedures required to copy OpenDDA to a Software Depot in preparation for installing OpenDDA onto your Application Module^x.

Preparing a Software Depot

The following table lists the steps for copying OpenDDA from a tape drive to a local HP-UX node that will be used as a software depot.

Table 3-1 Copying OpenDDA to a Software Depot

Step	Action
1	Log onto the system as root and enter the root password.
2	Install the appropriate software distribution tape in the DAT drive.
3	Enter the command: <code>/usr/sbin/swcopy</code> Press Enter.
4	A navigation display window appears explaining how to use the console terminal keys. Read the instructions, if necessary. Press Enter to continue.
5	A SD Copy - Software Selection (<i>hostname</i>) (1) window appears.
6	A Select Target Depot Path (<i>hostname</i>) window appears with the default path set to <code>/var/spool/sw</code> . Press the Tab key to go to OK and then press Enter to accept this default.
7	NOTE: You may get a screen telling you that the Target Depot Path and the Source Depot Path cannot be the same. If this happens, press Enter to continue.
8	A Specify Source (<i>hostname</i>) window appears with the entry in the Source Depot Type highlighted (selected). If the Source Depot Type selected is not Local Tape, press Enter to display additional options. Use the Up/Down Arrow keys to select Local Tape.

Table 3-1 Copying OpenDDA to a Software Depot

Step	Action
9	In most cases, the system will already have determined the correct path for the DAT drive (Local Tape) to be /dev/rmt/0m Press the Tab key to go to OK and then press Enter.
10	From the Options menu, select Change Options. Uncheck the Enforce Dependency Check here.
11	An SD Copy - Software Selection (hostname) (1) window appears that lists the software products available on the source tape. Mark the software you wish to copy with a Yes by performing the following: • Highlight the software product by pressing the space bar. • Press the Tab key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press Enter. • Use the down arrow key to select Mark for Copy from the Actions menu and then press Enter. The selected software should now be marked Yes.
12	An Error (hostname) window may appear if any of the software you selected has software dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies – it is just to let you know that they exist. Dependencies will be checked during software installation. Press Enter to return to the SD Copy – Software Selection window.
13	Repeat the above two steps for each software product that you wish to copy. When all the selections have been made, perform the following: • Press the Tab key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press Enter. • Use the down arrow key to select Copy Analysis from the Actions menu and then press Enter.
14	A Copy Analysis (hostname) (2) window appears displaying the Target, Status, and Products Scheduled information. When the analysis is complete, the Status indicates Ready or Ready with Errors (indicates that the program is unable to verify the software dependencies at this time). • If you wish to look at the logfile, select Logfile and press Enter. A Logfile (hostname) (2) window appears. Select OK to return to the Copy Analysis window. • Press the Tab key to go to OK and then press Enter.

Table 3-1 Copying OpenDDA to a Software Depot

Step	Action
15	A Confirmation (<i>hostname</i>) window appears. Select Yes and then press Enter.
16	A Copy Window (<i>hostname</i>) (3) window appears during the copy process displaying Target, Status, Percent Complete, Kbytes Installed, Time Left (minutes) and Copying Software information. The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Completed with Errors (indicates that the program is unable to verify the software dependencies).
17	Select Logfile and press Enter to display the copied software.
18	A Logfile (<i>hostname</i>) (2) window appears. Scroll through the log to determine if there were any errors during the copying process. Use the Tab key to go to OK and then press Enter to return to the Copy Window (<i>hostname</i>) (3) window.
19	Refer to the next section, Installing Other Software, when you are ready to install the copied software from the Software Depot source.
20	When you are ready to exit the copy program, perform the following: • Press the Tab key to go to DONE and press Enter. • Press the Tab key to go to the File selection of the menu bar and press Enter. • Use the down arrow key to select Exit and then press Enter.

3.3 Installing OpenDDA on the Local HP-UX Node

Procedure

The following table lists the steps for installing OpenDDA from a tape drive to a local HP-UX node.

Table 3-2 Installing OpenDDA on a Local HP-UX Node

Step	Action
1	Log onto the system as root and enter the root password.
2	Enter the command: <code>/usr/sbin/swinstall</code> Press Enter.
3	A navigation window appears explaining the use of the console terminal keys. If you are not familiar with the use of these keys, read the instructions. Press Enter to continue.
4	A window titled SD Install - Software Selection (<i>hostname</i>) (1) appears.
5	The install software will scan the default source depot type and will display a Note (<i>hostname</i>) window with a message displayed and with the OK selection highlighted. Press Enter to continue.
6	A Specify Source (<i>hostname</i>) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: • Press Enter to display additional options, if necessary - select local tape drive or software depot. • Use the Up/Down Arrow keys to select your software source. • Enter the correct path • Press the Tab key to go to OK and then press Enter.
7	A Bundles and Products window appears that lists the software products available on the specified source. Mark the software you wish to install with a Yes by performing the following: • Highlight the software product by pressing the space bar. • Press the Tab key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press Enter. • Use the down arrow key to select Mark for Install from the Actions menu and then press Enter. The selected software should now be marked Yes.

Table 3-2 Installing OpenDDA on a Local HP-UX Node

Step	Action
8	An Error (hostname) window may appear if any of the software you selected has software dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Press Enter to return to the Bundles and Products window.
9	Repeat the above two steps for each software product that you wish to install. When all the selections have been made, perform the following: • Press the Tab key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press Enter. • Use the down arrow key to select Install Analysis from the Actions menu and then press Enter.
10	An Install Analysis (hostname) (2) window appears with Target, Status, and Products Scheduled information displayed. The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready. If the Status does not indicate ready, you will need to correct the problem before continuing. Press the Tab key to go to OK and then press Enter.
11	A Confirmation (hostname) window appears. Select: Yes and press Enter.
12	Some software requires a system reboot after installation. If your system requires a reboot, another " Confirmation (hostname) " window will appear indicating that the system will be rebooted when installation is complete. If this occurs, Select: Yes and press Enter.
13	The " Install Window (hostname) (3) " window appears during the installation process displaying Target, Status, Percent Complete, Kbytes Installed, Time Left (minutes) and Loading Software information. The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready.
14	Press the Tab key to go to Logfile and then press Enter to display all the loaded modules.
15	A " Logfile (hostname) (2) " window displays. Scroll through the log to determine if there were any errors during the installation. Press the Tab key to go to OK and then press Enter.
16	The system will take you back to the " Install Window (hostname) (3) " window. Press the Tab key to go to DONE and then press Enter.
17	If your software requires a reboot, you will get a " Note (hostname) " window informing you that the system will be rebooted. Press Enter to continue.
18	If the software does not require a reboot, you will get the " SD Install - Software Selection (hostname) (1) " window. Press the Tab key to go to the menu bar (File will be selected) and then press Enter. Select "Exit" from the pull-down menu and then press Enter.

Files needed to link

When developing OpenDDA applications on a remote system, you will require the following files from the A^xM, in order to link your application and build an executable file.

- `/opt/TDC_Open/common/lib/libhiber.sl`
- `/opt/TDC_Open/common/lib/liblxs.sl`

The OpenDDA installation process will automatically load these files if they are not already there.

3.4 Directories and Files Created

Installation output

The following are the directories and files that are created when you load OpenDDA on the A^xM.

/opt/DDA

Directory : /opt/DDA

```
-rw-rw-rw- 1 axm axm .... DDACache
drwxr-xr-x 2 axm axm .... bin
drwxr-xr-x 2 axm axm .... doc
drwxr-xr-x 2 axm axm .... examples
drwxr-xr-x 2 axm axm .... include
drwxr-xr-x 2 axm axm .... lib
drwxr-xr-x 2 axm axm .... tmp
```

/opt/DDA/bin

Directory: /opt/DDA/bin

Files:

```
-rwxr-xr-x 1 axm axm .... c.c
-rwxr-xr-x 1 axm axm .... cmt
-r-xr-x--- 1 axm axm .... dda
-r-xr-xr-x 1 axm axm .... dda_system.def
-rwxr-xr-x 1 axm axm .... dda_template.def
-rwxr-xr-x 1 axm axm .... dda_xstatus
-rwxr-xr-x 1 axm axm .... fortran.f
```

/opt/DDA/lib

Directory: /opt/DDA/lib

Files:

```
-r-xr-xr-x 1 axm axm .... libdda.sl
-r-xr-xr-x 1 axm axm .... libddagdx.sl
-r-xr-xr-x 1 axm axm .... libddagen.sl
```

/opt/DDA/include

Directory: /opt/DDA/include

Files:

```
-rwxr-xr-x 1 axm axm .... dda_c.h
-rwxr-xr-x 1 axm axm .... dda_c_errors.h
-rwxr-xr-x 1 axm axm .... dda_fortran.f
-rwxr-xr-x 1 axm axm .... dda_fortran_errors.f
-rwxr-xr-x 1 axm axm .... dda_status_include.h
-rwxr-xr-x 1 axm axm .... hgsc_c_errors.h
-rwxr-xr-x 1 axm axm .... hgsc_fortran_errors.f
```

/opt/DDA/examples

Directory: /opt/DDA/examples

Files:

```
-rwxr-xr-x 1 axm axm .... alltestc.c
-rwxr-xr-x 1 axm axm .... alltestc.def
-rwxr-xr-x 1 axm axm .... alltestf.def
-rwxr-xr-x 1 axm axm .... alltestf.f
-rwxr-xr-x 1 axm axm .... dda_c_examples.make
-rwxr-xr-x 1 axm axm .... dda_create_c_debug_executable
-rwxr-xr-x 1 axm axm .... dda_create_c_executable
-rwxr-xr-x 1 axm axm .... dda_create_cpp_debug_executable
-rwxr-xr-x 1 axm axm .... dda_create_cpp_executable
-rwxr-xr-x 1 axm axm .... dda_create_fortran_debug_executable
-rwxr-xr-x 1 axm axm .... dda_create_fortran_executable
-rwxr-xr-x 1 axm axm .... ddademo.c
-rwxr-xr-x 1 axm axm .... ddademo.def
-rwxr-xr-x 1 axm axm .... ddademo.f
-rwxr-xr-x 1 axm axm .... dda_fortran_examples.make
-rwxr-xr-x 1 axm axm .... lvr.f.def
-rwxr-xr-x 1 axm axm .... event_hiber.c
-rwxr-xr-x 1 axm axm .... readit.def
-rwxr-xr-x 1 axm axm .... readit.f
-rwxr-xr-x 1 axm axm .... read_tank_info.c
-rwxr-xr-x 1 axm axm .... remap.cf
-rwxr-xr-x 1 axm axm .... remap2.edb
-rwxr-xr-x 1 axm axm .... remap3.edb
```

/opt/DDA/doc

Directory: /opt/DDA/doc

File:

-rwxrwxrwx 1 axm axm size date dda.help

/opt/DDA/tmp

Temporary work directory used by OpenDDA.

3.5 Loading OpenDDA on a Workstation

Discussion

When you are loading OpenDDA on a workstation for application development, all of the directories and files covered in subsection 3.4 will be loaded. In addition, there is a directory with two library files that are standard in an A^xM that you must have. These two files are required for linking an application.

The OpenDDA installation process checks for the existence of the directory and three library files, and if they are not present, the directory is created and the files are loaded.

/opt/TDC_Open/ common/lib

The directory `/opt/TDC_Open/common/lib` will be created if not already present. The following files will be loaded into this directory if not already present:

```
-r-xr-xr-x 1 root bin .... libhiber.sl  
-r-xr-xr-x 1 root bin .... liblxs.sl
```

3.6 Adding Users

Assigning axm group

When creating OpenDDA users, be sure to assign each user as a member of the "axm" Group.

Attention

The user must be a member of the "axm" group to install and execute an OpenDDA application.

Reference

Refer to subsection 4.3 Adding and Removing Users in the *Application Module^x System Administration* manual for instructions on adding OpenDDA users.

3.7 Migrating OpenDDA R100 and R110 Applications

Loading software

To migrate your A^xM to OpenDDA R300, it is important to load software in a specific order. Refer to Software loading order in the Special Considerations Before You Install subsection of this manual.

Installed CL- initiated applications

No actions are required for OpenDDA applications installed as CL- initiated tasks in the /users/axm directory. After all software is loaded, these applications will execute when the system is brought back up.

Script and make file changes

Because of necessary library name changes, additional libraries to include, and OpenDDA FORTRAN source file name change, the example script files and make files have changed in the /opt/DDA/examples directory. If you have copied and modified these files, you will need to review the new files and make any necessary changes to your custom script and make files.

Changes to applications

If any changes are made to your application, you will be required to fully rebuild your OpenDDA application. The following steps should be taken after an application has been modified, parsed, or even relinked.

Table 3-3 Steps to Rebuild an OpenDDA Application

Step	Action
1	Use the Complete Build option (-cb) with the dda command to recreate the OpenDDA .set, .set_idx, and .cio file.
2	Use the appropriate example script file provided in the /opt/DDA/examples directory to compile and link your application.
3	Execute the application to verify functionality.

4. Problem Reporting

4.1 Overview

Introduction

The information provided in this section is intended to assist the administrator and Honeywell's Technical Assistance Center (TAC) in resolving problems encountered when developing and executing applications in OpenDDA.

4.2 Errors While Building an Application

Build application errors

While using any of the command line features in OpenDDA, the most important information is output to the screen when using these functions. The following is a list of files Honeywell's Technical Assistance Center (TAC) may require when determining the resolution to your problem:

- the Application's Definition File (.def)
- the application's file containing both the EDB statements and EXEC DDA statements.

Additional information

Additional information for TAC may be any or all of the following:

- command in use when the error occurs
- a screen dump of the output generated by OpenDDA when the error occurred
- a snapshot of the current processes running on the system
- any script files or make files used when executing the OpenDDA command
- release number of the OpenDDA software and A^xM libraries—you can determine these with the commands:

```
what /opt/DDA/bin/dda
```

```
what /opt/TDC_Open/common/lib/*.sl
```

- the application's .edb (External Data Block) file
- the checksum value of the shared libraries on your system—you can determine these with the commands:

```
sum -p /opt/DDA/lib/*.sl
```

```
sum -p /opt/TDC_Open/common/lib/*.sl
```

4.3 Execution Errors

Execution output file

OpenDDA creates an output file when it executes an application. The file name has the format:

`your_application_name.dda_out`

This file can contain error and information messages that will assist in debugging your applications.

Output file contents

The **.dda_out** file may contain the following information:

- output generated by the PRINT switches set in your application's Definition file (**.def**)
- output from the application itself if you have written to Standard Output
- error or warning messages from DDA if your application experienced problems during execution.

Attention

Be sure you turn ON the PRINT Values, Statistics, and Messages in the application's Definition file before executing the application, when you are debugging your application.

5. Known Problems

5.1 Outstanding PARs and Known Problems

Enumerated index arrays

OpenDDA cannot access arrays indexed by enumerations.

Reading an array's entity_id parameters

If the entity_id your OpenDDA application is accessing is NULL, the value returned to your application depends on the value status. If the status is good, then your value will be a blank string. If the value status is not good, then the value will be the previous value accessed.

TEST variable problem

The TEST ancillary variable does not function properly with the use of multiple test values and FULL_NAME alias arrays.

The use of the TEST ancillary variable with multiple test values and FULL_NAME alias arrays results in invalid run time read access for the TEST ancillary variable. This only affects the HLL value variable that is associated with the TEST ancillary keyword in the EDB. Only the first test value will be correctly populated into the HLL variable as opposed to all of the test values.

Example:

```
val1[0:3] = 1.0, 2.0, 3.0, 4.0 /REAL
val1_sts[0:3] = INPUT CURRENT (, *.*(*))
val1_fn[0:3] = FULL_NAME
val1_testval[0:3] = TEST
SET_LIST = SET_A;
```

When SET_A is read, only the first element of the val1_testval array will be populated correctly with the value, 1.0. The values populated for the remaining three array elements will be undefined.

⇒ NOTE: The population of 'live' data into val1 is unaffected.

Workaround:

By using a single test value, proper functionality can be achieved.

TEST MODE problem

The non-sequential duplication of the first test value in a list of test values results in invalid test mode read access.

If multiple test values are used where the first test value is repeated non-sequentially in the list, then improper test mode access will result such that invalid test values are populated.

Example:

```
val1[0:3] = 1.0, 2.0, 1.0, 4.0 /REAL  
val1_sts[0:3] = INPUT CURRENT (, PT1.PARAM1(1))  
SET_LIST = SET_A;
```

Because the first test value, 1.0, is duplicated as the third test value (non-sequential duplication), upon reading SET_A improper values will be returned into the val1 array when executing in test mode.

NOTE: Normal 'live' data access is unaffected.

Workaround:

Make sure the first test value in the list is unique or do not use test mode.

5.2 Problems Fixed in the R200 Release

List of fixed problems

The following PAR and other problems were fixed as part of the R200 release:

- Accessing the zeroth element of an array on the LCN
- Accessing individual elements of arrays greater than the 257th element on the LCN
- The size of OpenDDA's .set file has been cut in half
- The amount of memory consumed by OpenDDA for processing numerous SETs declared in the application has been reduced
- When the values print switch was turned on in the application's .def file, and the user had a STORE_STATUS defined in the EDB, the STORE_STATUS value was being printed during an EXEC DDA READ command, causing a false error message to be printed in the .dda_output file.

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